Arab Academy for Science, Technology & Maritime Transport
College of Engineering and Technology
Final Examination Paper

Department: Computer Engineering
Course Title: Systems Programming
Course Code: CC410
Lecturer: Dr. Sherine Nagy Saleh

Date : 3 February 2021
Time : **2** Hours
Marks: 40

<u>Attempt all questions. Show all your work so that partial credit may be awarded</u>

<u>Question # 1 [ILO A1][OUTCOME 2]:</u> (10 Marks)

a. (4 marks) Disassemble the following program

```
HTEST  000000000019
T00000013010003132010050000 6F10001690010F900013
E000000
```

b. (6 marks) Assume the following SIC/XE code:

0030 VAR1 BYTE 0123H

0200 VAR2 BYTE 0080H

For each of the following instructions ***explain*** what happens when each is ***executed***. Generate ***machine code*** for each of the instructions and in each case state the ***addressing mode*** and the ***format*** of the instruction which has been used.

i. **0020 LDA VAR1**
ii. **0020 LDA @VAR1**
iii. **0020 LDA #5**
iv. **0020 +LDA VAR2**

Question # 2 [ILO A1][OUTCOME 2]: (10 Marks)

Given the following SIC program:

a. **(3 marks)** Generate the complete H T E records showing all your steps

b. **(4 marks)** Using an absolute loader, show the memory content after the object code is loaded into the memory

c. **(3 marks)** In order to make the program relocatable, a system software programmer decided to use modification records. Generate the modification records needed for this program.

TEST	START	1000
BEGIN	LDA	GAMMA
	SUB	FIRST
	STA	RESULT
	LDA	ALPHA
	ADD	BETA
	SUB	RESULT
	STA	RESULT
ALPHA	WORD	X'48934D'
BETA	WORD	X'123456'
GAMMA	BYTE	X'4C',X'9E',X'36'
FIRST	BYTE	X'12',X'65',X'78'
RESULT	RESW	1
	END	

Question # 3 [ILO A1][OUTCOME 2]: (10 Marks)

Given the following two control sections:

EXAM	START	0
	EXTDEF	LISTA, ENDA, BUFFER, BUFEND
	EXTREF	LISTB, ENDB
REF1	+LDA	#MAXLEN
REF2	WORD	ENDB-LISTB+LISTA-3
REF3	WORD	ENDB-LISTA
REF4	+LDA	LISTB
REF5	RESW	3
BUFFER	REBB	4096
BUFEND	EQU	*
LISTA	WORD	3
ENDA	RESW	1
MAXLEN	EQU	BUFEND-BUFFER

SUB1	CSECT	
	EXTDEF	LISTB, ENDB
	EXTREF	LISTA, ENDA, BUFFER, BUFEND
REF1	LDA	MAXLEN, X
REF2	+LDA	LISTA
REF3	RESW	2
REF4	WORD	ENDB-LISTB+LISTA-3
REF5	WORD	ENDB-ENDA
TEMP	RESW	2
LISTB	WORD	1
ENDB	EQU	*
MAXLEN	WORD	BUFEND-BUFFER
	END	REF1

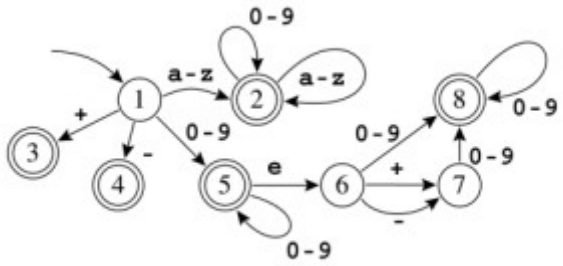
a. **(5 marks)** Generate the complete H T E for ***each*** control section showing all your steps

b. **(3 marks)** Assume EXAM is loaded at address 4000 and SUB1 is loaded at address 4100. Generate the external symbol table.

c. **(2 marks)** Show the address and the memory contents ***only*** of the lines that will be modified at linking.

Question # 4 [ILO A1][OUTCOME 2]: (10 Marks)

a. **(3 marks)** Using the following transition diagram**, suggest** 6 different tokens that can be identified. Each suggested token has to pass through one of states 2, 3, 4, 5 and 8 (all the mentioned states have to be used). Also the tokens have to show variety in length (if applicable)



b. **(3 marks)** For the following two regular definitions:

def1 -> a(a|b)*a def2-> a*ba*ba*ba*

- Describe the languages denoted by the regular definitions.
- Suggest two different token that could be detected by each def1 and def2

c. **(4 marks)** For the grammar shown here, derive the parse tree for the following expression

2*(3+5*4)

Grammar:
E→EAE
E→(E)
E→-E
E->**number**
A→+
A→-
A→+
A→/

Best Wishes ☺

Mnemonic	Format	Opcode	Effect
ADD m	3/4	18	A ← (A) + (m..m+2)
ADDR r1,r2	2	90	r2->r2+r1
AND m	3/4	40	A ← (A) & (m..m+2)
CLEAR r1	2	B4	r1 ← 0
COMP m	3/4	28	A : (m..m+2)
DIV m	3/4	24	A : (A) / (m..m+2)
HIO	1	F4	Halt I/O channel number (A)
J m	3/4	3C	PC ← m
JEQ m	3/4	30	PC ← m if CC set to =
JGT m	3/4	34	PC ← m if CC set to >
JLT m	3/4	38	PC ← m if CC set to <
JSUB m	3/4	48	L ← (PC); PC ← m
LDA m	3/4	00	A ← (m..m+2)
LDCH m	3/4	50	A [rightmost byte] ← (m)
LDB m	3/4	68	B ← (m..m+2)
LDL m	3/4	08	L ← (m..m+2)
LDS m	3/4	6C	S ← (m..m+2)
LDT m	3/4	74	T ← (m..m+2)
LDX m	3/4	04	X ← (m..m+2)
MUL m	3/4	20	A ← (A) * (m..m+2)
OR m	3/4	44	A ← (A) | (m..m+2)
RD m	3/4	D8	A [rightmost byte] ← data
RSUB	3/4	4C	PC ← (L)
STA m	3/4	0C	m..m+2 ← (A)
STCH m	3/4	54	m ← (A) [rightmost byte]
STL m	3/4	14	m..m+2 ← (L)
STSW m	3/4	E8	m..m+2 ← (SW)
STX m	3/4	10	m..m+2 ← (X)
SUB m	3/4	1C	A ← (A) - (m..m+2)
TD m	3/4	E0	Test device specified by (m)
TIO	1	F8	Test I/O channel number (A)
TIX m	3/4	2C	X ← (X) + 1; (X) : (m..m+2)
TIXR r1	2	B8	X←(X)+1; (X): (r1)
WD m	3/4	DC	Device specified by (m) ← (A) [rightmost byte]

Register Name	ID
A	0
X	1
L	2
B	3
S	4
T	5
F	6